

DAY THREE

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Scalar and Vector

Learning & Revision for the Day

- Scalar and Vector Quantities
- Laws of Vector Addition
- Subtraction of Vectors
- Multiplication or Division of a Vector by a Scalar
- Product of Vectors
- Resolution of vector
- Relative velocity
- Motion in a Plane
- Projectile Motion

Scalar and Vector Quantities

A **scalar quantity** is one whose specification is completed with its magnitude only. e.g. mass, distance, speed, energy, etc.

A **vector quantity** is a quantity that has magnitude as well as direction. Not all physical quantities have a direction. e.g. velocity, displacement, force, etc.

Position and Displacement Vectors

A vector which gives position of an object with reference to the origin of a coordinate system is called position vector.

The vector which tells how much and in which direction an object has changed its position in a given interval of time is called displacement vector.

General Vectors and Notation

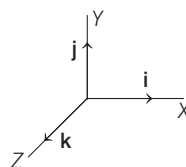
- **Zero Vector** The vector having zero magnitude is called **zero vector** or **null vector**. It is written as 0. The initial and final points of a zero vector overlap, so its direction is arbitrary (not known to us).
- **Unit Vector** A vector of unit magnitude is known as an **unit vector**. Unit vector for **A** is $\hat{A}$ (read as A cap).

$$\mathbf{A} = A \hat{\mathbf{A}} \quad \text{Direction}$$

|
Magnitude

- **Orthogonal Unit Vectors** The unit vectors along X-axis, Y-axis and Z-axis are denoted by $\hat{i}$, $\hat{j}$ and $\hat{k}$. These are the orthogonal unit vectors.

$$\hat{i} = \frac{x}{x}, \hat{j} = \frac{y}{y}, \hat{k} = \frac{z}{z}$$



- **Parallel Vector** Two vectors are said to be parallel, if they have same direction but their magnitudes may or may not be equal.
- **Antiparallel Vector** Two vectors are said to be anti-parallel when
 - (i) both have opposite direction
 - (ii) one vector is scalar non zero negative multiple of another vector.
- **Collinear Vector** Collinear vector are those which act along same line.
- **Coplanar Vector** Vector which lies on the same plane are called coplanar vector.
- **Equal Vectors** Two vectors **A** and **B** are equal, if they have the same magnitude and the same direction.

Laws of Vector Addition

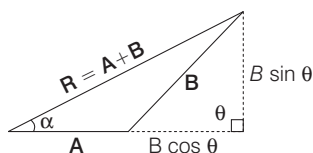
1. Triangle Law

If two non-zero vectors are represented by the two sides of a triangle taken in same order then the resultant is given by the closing side of triangle in opposite order, i.e.

$$\mathbf{R} = \mathbf{A} + \mathbf{B}$$

The resultant **R** can be calculated as

$$|\mathbf{A} + \mathbf{B}| = R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$



If resultant **R** makes an angle α with vector **A**, then

$$\tan \alpha = \frac{B \sin \theta}{A + B \cos \theta}$$

2. Parallelogram Law

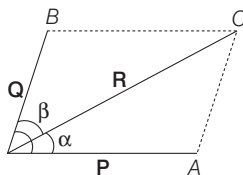
According to parallelogram law of vector addition, if two vector acting on a particle are represented in magnitude and direction by two adjacent side of a parallelogram, then the diagonal of the parallelogram represents the magnitude and direction of the resultant of the two vector acting as the particle.

$$\text{i.e. } \mathbf{R} = \mathbf{P} + \mathbf{Q}$$

Magnitude of the resultant **R** is given by

$$|\mathbf{R}| = \sqrt{P^2 + Q^2 + 2PQ \cos \theta}$$

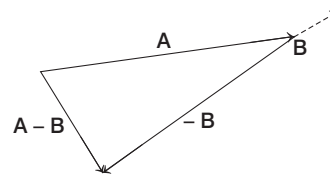
$$\tan \alpha = \frac{Q \sin \theta}{P + Q \cos \theta} \Rightarrow \tan \beta = \frac{P \sin \theta}{Q + P \cos \theta}$$



Subtraction of Vectors

Vector subtraction makes use of the definition of the negative of a vector. We define the operation $\mathbf{A} - \mathbf{B}$ as vector $-\mathbf{B}$ added to vector **A**. $\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B})$

Thus, vector subtraction is really a special case of vector addition. The geometric construction for subtracting two vectors is shown in the above figure.



If θ be the angle between **A** and **B**,

$$\text{then } |\mathbf{A} - \mathbf{B}| = \sqrt{A^2 + B^2 - 2AB \cos \theta}$$

If the vectors form a closed n sided polygon with all the sides in the same order, then the resultant is zero.

Multiplication or Division of a Vector by a Scalar

The multiplication or division of a vector by a scalar gives a vector. For example, if vector **A** is multiplied by the scalar number 3, the result, written as $3\mathbf{A}$, is a vector with a magnitude three times that of **A**, pointing in the same direction as **A**. If we multiply vector **A** by the scalar -3 , the result is $-3\mathbf{A}$, a vector with a magnitude three times that of **A**, pointing in the direction opposite to **A** (because of the negative sign).

Product of Vectors

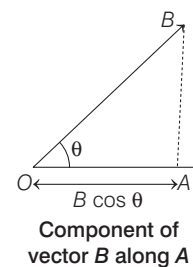
The two types of products of vectors are given below

Scalar or Dot Product

The scalar product of two vectors **A** and **B** is defined as the product of magnitudes of **A** and **B** multiplied by the cosine of smaller angle between them. i.e. $\mathbf{A} \cdot \mathbf{B} = AB \cos \theta$

Properties of Dot Product

- Dot product or scalar product of two vectors gives the scalar two vectors given the scalar quantity.
- It is commutative in nature. i.e. $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$.
- Dot product is distributive over the addition of vectors. i.e. $\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}$
- $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$, because angle between two equal vectors is zero.
- If two vectors **A** and **B** are perpendicular vectors, then $\mathbf{A} \cdot \mathbf{B} = AB \cos 90^\circ = 0$ and $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$



The Vector Product

The vector product of **A** and **B**, written as $\mathbf{A} \times \mathbf{B}$, produces a third vector **C** whose magnitude is $C = AB \sin \theta$. where, θ is the smaller of the two angles between **A** and **B**.

Because of the notation, $\mathbf{A} \times \mathbf{B}$ is also known as the **cross product**, and it is spelled as 'A cross B'.

Properties of Cross Product

- Vector or cross product of two vectors gives the vector quantity.
- Cross product of two vectors does not obey the commutative law. i.e. $\mathbf{A} \times \mathbf{B} \neq \mathbf{B} \times \mathbf{A}$;
Here, $\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}$
- Cross product of two vectors is distributive over the addition of vectors.

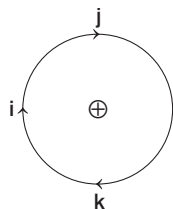
$$\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = \mathbf{A} \times \mathbf{B} + \mathbf{A} \times \mathbf{C}$$

- Cross product of two equal vectors is given by $\mathbf{A} \times \mathbf{A} = 0$
Similarly, $\hat{\mathbf{i}} \times \hat{\mathbf{i}} = (1 \times 1 \times \sin 0^\circ) \hat{\mathbf{n}} = 0$
 $\hat{\mathbf{j}} \times \hat{\mathbf{j}} = (1 \times 1 \times \sin 0^\circ) \hat{\mathbf{n}} = 0$
 $\hat{\mathbf{k}} \times \hat{\mathbf{k}} = (1 \times 1 \times \sin 0^\circ) \hat{\mathbf{n}} = 0$

- Cross product of two perpendicular vectors is given as $\mathbf{A} \times \mathbf{B} = (AB \sin 90^\circ) \hat{\mathbf{n}} = (AB) \hat{\mathbf{n}}$
- For two vectors $\mathbf{A} = a_x \hat{\mathbf{i}} + a_y \hat{\mathbf{j}} + a_z \hat{\mathbf{k}}$
and $\mathbf{B} = b_x \hat{\mathbf{i}} + b_y \hat{\mathbf{j}} + b_z \hat{\mathbf{k}}$.

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$$

- Cross product of vectors $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$ and $\hat{\mathbf{k}}$ are following cyclic rules as follows $\hat{\mathbf{i}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}}$, $\hat{\mathbf{j}} \times \hat{\mathbf{k}} = \hat{\mathbf{i}}$ and $\hat{\mathbf{k}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}}$



Cyclic representation for unit vectors $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$ and $\hat{\mathbf{k}}$

NOTE

- Vector triple product is given by $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B} (\mathbf{A} \cdot \mathbf{C}) - \mathbf{C} (\mathbf{A} \cdot \mathbf{B})$

Resolution of a Vector

The process of splitting of a single vector into two or more vectors in different direction is resolution of a vectors. Consider a vector $\mathbf{A}$ in the X - Y plane making an angle θ with the X -axis. The X and Y components of $\mathbf{A}$ are A_x and A_y respectively.

$$\text{Thus } \mathbf{A}_x = \mathbf{A}_{xi} = (A \cos \theta) \hat{\mathbf{i}}$$

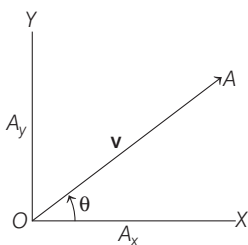
along X -direction

$$\mathbf{A}_y = \mathbf{A}_{yj} = (A \sin \theta) \hat{\mathbf{j}} \text{ along } Y\text{-direction}$$

From triangle law of vector addition

$$|\mathbf{A}| = |\mathbf{A}_{xi} + \mathbf{A}_{yj}| = \sqrt{A_x^2 + A_y^2}$$

$$\text{and } \tan \theta = \frac{A_y}{A_x} = \theta = \tan^{-1} \left(\frac{A_y}{A_x} \right)$$



Relative Velocity

The time rate of change of relative position of one object with respect to another is called relative velocity.

Different Cases

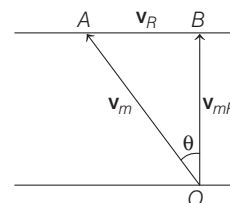
Case I If both objects A and B move along parallel straight lines in the opposite direction, then relative velocity of B w.r.t. A is given as,

$$\mathbf{v}_{BA} = \mathbf{v}_B - (-\mathbf{v}_A) = \mathbf{v}_B + \mathbf{v}_A$$

If both objects A and B move along parallel straight lines in the same direction, then

$$\mathbf{v}_{AB} = \mathbf{v}_B - \mathbf{v}_A$$

Case II Crossing the River To cross the river over shortest distance, i.e. to cross the river straight, the man should swim upstream making an angle θ with $\mathbf{OB}$ such that, $\mathbf{OB}$ gives the direction of resultant velocity ($\mathbf{v}_{mR}$) of velocity of swimmer $\mathbf{v}_m$ and velocity of river water $\mathbf{v}_R$ as shown in figure. Let us consider



$$\text{In } \triangle OAB, \sin \theta = \frac{v_R}{v_m} \text{ and } v_{mR} = \sqrt{v_m^2 - v_R^2}$$

The time taken to cross the river given by

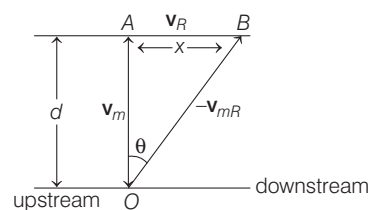
$$t_1 = \frac{d}{v_{mR}} = \frac{d}{\sqrt{v_m^2 - v_R^2}}$$

Case III To cross the river in possible shortest time The man should go along OA . Now, the swimmer will be going along OB , which is the direction of resultant velocity $\mathbf{v}_{mR}$ of $\mathbf{v}_m$ and $\mathbf{v}_R$.

$$\text{In } \triangle OAB, \tan \theta = \frac{AB}{OA} = \frac{v_R}{v_m}$$

and

$$v_{mR} = \sqrt{v_m^2 + v_R^2}$$



Time of crossing the river,

$$t = \frac{d}{v_m} = \frac{OB}{v_{mR}} = \frac{\sqrt{x^2 + d^2}}{\sqrt{v_m^2 + v_R^2}}$$

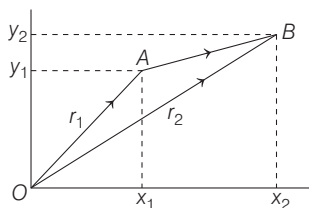
The boat will be reaching the point B instead of point A . If $AB = x$,

$$\text{then, } \tan \theta = \frac{v_R}{v_m} = \frac{x}{d} \Rightarrow x = \frac{dv_R}{v_m}$$

Motion in a Plane

Let the object be at position A and B at timing t_1 and t_2 , where $OA = r_1$, and $OB = r_2$

Suppose O be the origin for measuring time and position of the object (see figure).



- Displacement of an object from position A to B is

$$AB = r = r_2 - r_1 = (x_2 - x_1)\hat{i} - (y_2 - y_1)\hat{j}$$

- Velocity, $v = \frac{r_2 - r_1}{t_2 - t_1}$

- A particle moving in X - Y plane (with uniform velocity) then, its equation of motion for X and Y axes are

$$v = v_x\hat{i} + v_y\hat{j}, r_0 = x_0\hat{i} + y_0\hat{j} \text{ and } r = x\hat{i} + y\hat{j}$$

$$x = x_0 + v_x t, y = y_0 + v_y t$$

- A particle moving in xy -plane (with uniform acceleration), then its equation of motion for X and Y -axes are

$$v_x = u_x + a_x t, v_y = u_y + a_y t$$

$$x = x_0 + u_x t + \frac{1}{2} a_x t^2, y = y_0 + u_y t + \frac{1}{2} a_y t^2$$

$$a = a_x\hat{i} + a_y\hat{j}$$

Projectile Motion

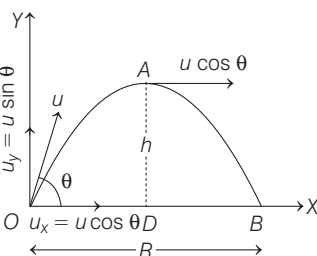
Projectile is an object which once projected in a given direction with given velocity and is then free to move under gravity alone. The path described by the projectile is called its trajectory.

Let a particle is projected at an angle θ from the ground with initial velocity u .

Resolving u in two components, we have

$$u_x = u \cos \theta, u_y = u \sin \theta, a_x = 0, a_y = -g.$$

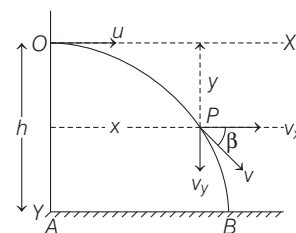
- Equation of trajectory, $y = x \tan \theta - \frac{g}{2u^2 \cos^2 \theta} x^2$
- Vertical height covered, $h = \frac{u^2 \sin^2 \theta}{2g}$
- Horizontal range, $R = OB = u_x T, R = \frac{u^2 \sin 2\theta}{g}$



Projectile Motion in Horizontal Direction From Height (h)

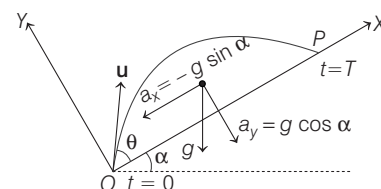
Let a particle be projected in horizontal direction with speed u from height h .

- Equation of trajectory, $y = \frac{gx^2}{2u^2}$
- Time of flight, $T = \frac{\sqrt{2h}}{g}$
- Horizontal range, $R = u \sqrt{\frac{2h}{g}}$
- Velocity of projectile at any time, $v = \sqrt{u^2 + g^2 t^2}$



Projectile Motion Up an Inclined Plane

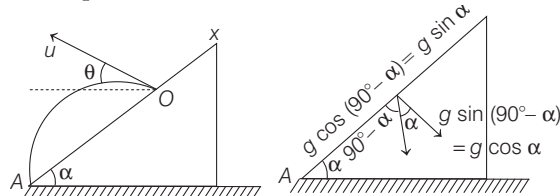
Let a particle be projected up with speed u from an inclined plane which makes an angle α with the horizontal and velocity of projection makes an angle θ with the inclined plane.



- Time of flight on an inclined plane $T = \frac{2u \sin \theta}{g \sin \alpha}$
- Maximum height, $h = \frac{u^2 \sin^2 \theta}{2g \cos \alpha}$
- Horizontal range, $R = \frac{2u^2 \sin \theta \cos (\theta + \alpha)}{g \cos^2 \alpha}$
- Maximum range occurs when $\theta = \frac{\pi}{2} - \frac{\alpha}{2}$
- $R_{\max} = \frac{u^2}{g(1 + \sin \alpha)}$ when projectile is thrown upwards.
- $R_{\max} = \frac{u^2}{g(1 - \sin \alpha)}$ when projectile is thrown downwards.

Projectile Motion Down an Inclined Plane

A projectile is projected down the plane from the point O with an initial velocity u at an angle θ with horizontal. The angle of inclination of plane with horizontal α . Then,



- Time of flight down an inclined plane, $T = \frac{2u \sin (\theta + \alpha)}{g \cos \alpha}$
- Horizontal range, $R = \frac{u^2}{g \cos^2 \alpha} [\sin (2\theta + \alpha) + \sin \alpha]$

DAY PRACTICE SESSION 1

FOUNDATION QUESTIONS EXERCISE

- Which of the following statement is true?
 - A scalar quantity is the one that is conserved in a process
 - A scalar quantity is one that can never be negative values
 - A scalar quantity is the one that does not vary from one point to another in space
 - A scalar quantity has the same value for observers with different orientations of the axes
- If two vectors are equal in magnitude and their resultant is also equal in magnitude to one of them, then the angle between the two vectors is
 - 60°
 - 120°
 - 90°
 - 0°
- If $\mathbf{A} = 3\hat{i} + 4\hat{j}$ and $\mathbf{B} = 7\hat{i} + 24\hat{j}$, the vector having the same magnitude as $\mathbf{B}$ and parallel to $\mathbf{A}$ is
 - $5\hat{i} + 20\hat{j}$
 - $15\hat{i} + 10\hat{j}$
 - $20\hat{i} + 15\hat{j}$
 - $15\hat{i} + 20\hat{j}$
- Six vectors $\mathbf{a}$ through $\mathbf{f}$ have the magnitudes and directions as shown in figure. Which statement is true?

→ CBSE AIPMT 2010

$\begin{array}{c} \vec{a} \\ \uparrow \end{array}$

$\begin{array}{c} \downarrow \vec{b} \\ \vec{e} \end{array}$

$\begin{array}{c} \vec{c} \\ \searrow \end{array}$

 - $\mathbf{b} + \mathbf{c} = \mathbf{f}$
 - $\mathbf{d} + \mathbf{c} = \mathbf{f}$
 - $\mathbf{a} + \mathbf{e} = \mathbf{f}$
 - $\mathbf{b} + \mathbf{e} = \mathbf{f}$
- The component of vector $\mathbf{A} = 2\hat{i} + 3\hat{j}$ along the vector $\hat{i} + \hat{j}$ is
 - $\frac{5}{\sqrt{2}}$
 - $10\sqrt{2}$
 - $5\sqrt{2}$
 - 5
- $\mathbf{A}$ and $\mathbf{B}$ are two vectors and θ is the angle between them, if $|\mathbf{A} \times \mathbf{B}| = \sqrt{3} (\mathbf{A} \cdot \mathbf{B})$, the value of θ is
 - 60°
 - 45°
 - 30°
 - 90°
- Given $\mathbf{A} = 4\hat{i} + 6\hat{j}$ and $\mathbf{B} = 2\hat{i} + 3\hat{j}$. Which of the following is correct?
 - $\mathbf{A} \times \mathbf{B} = 0$
 - $\mathbf{A} \cdot \mathbf{B} = 24$
 - $\frac{|\mathbf{A}|}{|\mathbf{B}|} = \frac{1}{2}$
 - $\mathbf{A}$ and $\mathbf{B}$ are anti-parallel
- If $\mathbf{A} = 4\hat{i} + 4\hat{j} + 4\hat{k}$ and $\mathbf{B} = 3\hat{i} + \hat{j} + 4\hat{k}$, then angle between vectors $\mathbf{A}$ and $\mathbf{B}$ is
 - 180°
 - 90°
 - 45°
 - 0°
- If two vectors $2\hat{i} + 3\hat{j} + \hat{k}$ and $-4\hat{i} - 6\hat{j} - \lambda\hat{k}$ are parallel to each other, then value of λ is
 - zero
 - 2
 - 3
 - 4
- If $\mathbf{A} \cdot \mathbf{B} = \mathbf{A} \times \mathbf{B}$, then the angle between $\mathbf{A}$ and $\mathbf{B}$ is
 - 45°
 - 30°
 - 60°
 - 90°
- If a vector $2\hat{i} + 3\hat{j} + 8\hat{k}$ is perpendicular to the vector $4\hat{j} - 4\hat{i} + \alpha\hat{k}$, then value of α is
 - 1
 - $\frac{1}{2}$
 - $-\frac{1}{2}$
 - 1
- At what angle should the two forces $2P$ and $\sqrt{2}P$ act, so that the resultant force is $P\sqrt{10}$?
 - 45°
 - 60°
 - 90°
 - 120°
- A boat is sent across a river with a velocity of 8 km/h. If the resultant velocity of boat is 10 km/h, then velocity of river is
 - 10 km/h
 - 8 km/h
 - 6 km/h
 - 4 km/h
- The velocity of a projectile at the initial point A is $(2\hat{i} + 3\hat{j})$ m/s. Its velocity (in m/s) at point B is

→ NEET 2013

 - $-2\hat{i} - 3\hat{j}$
 - $-2\hat{i} + 3\hat{j}$
 - $2\hat{i} - 3\hat{j}$
 - $2\hat{i} + 3\hat{j}$
- The x and y coordinates of the particle at any time are $x = 5t - 2t^2$ and $y = 10t$ respectively, where x and y are in metres and t in seconds. The acceleration of the particle at $t = 2$ s is

→ NEET 2017

 - 0
 - $5\hat{i} \text{ m/s}^2$
 - $-4\hat{i} \text{ m/s}^2$
 - $-8\hat{i} \text{ m/s}^2$
- A particle has initial velocity $(3\hat{i} + 4\hat{j})$ and has acceleration $(0.4\hat{i} + 0.3\hat{j})$. Its speed after 10 s is

→ CBSE AIPMT 2010

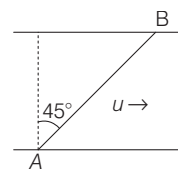
 - 7 unit
 - $7\sqrt{2}$ unit
 - 8.5 unit
 - 10 unit
- A particle is moving such that its position coordinates (x, y) are (2 m, 3 m) at time $t = 0$, (6 m, 7 m) at time $t = 2$ s and (13 m, 14 m) at time $t = 5$ s. Average velocity vector ($\mathbf{v}_{av}$) from $t = 0$ to $t = 5$ s is
 - $\frac{1}{5} (13\hat{i} + 14\hat{j})$
 - $\frac{7}{3} (\hat{i} + \hat{j})$
 - $2 (\hat{i} + \hat{j})$
 - $\frac{11}{5} (\hat{i} + \hat{j})$
- The horizontal range and maximum height attained by a projectile are R and H , respectively. If a constant horizontal acceleration $a = g/4$ is imparted to the projectile due to wind, then its horizontal range and maximum height will be
 - $(R + H), \frac{H}{2}$
 - $\left(R + \frac{H}{2}\right), 2H$
 - $(R + 2H), H$
 - $(R + H), H$

- 19** A projectile is thrown in the upward direction making an angle of 60° with the horizontal direction with a velocity of 147 m/s. Then, the time after which its inclination with the horizontal is 45° , is
 (a) 15 s (b) 10.98 s (c) 5.49 s (d) 2.745 s
- 20** The velocity of a particle is $v = v_0 + gt + at^3$. If its position is $x = 0$ at $t = 0$, then its displacement after unit time ($t = 1$) is
 (a) $v_0 = \frac{g}{2} + a$ (b) $v_0 = 2g + 3a$
 (c) $v_0 = \frac{g}{2} + \frac{a}{3}$ (d) $v_0 = g + a$
- 21** A projectile is fired from the surface of the earth with a velocity of 5 m/s and angle θ with the horizontal. Another projectile fired from another planet with a velocity of 3 m/s at the same angle follows a trajectory which is identical with the trajectory of the projectile fired from the earth. The value of the acceleration due to gravity on the planet is (in m/s^2) (given, $g = 9.8 \text{ m/s}^2$) → CBSE AIPMT 2014
 (a) 3.5 (b) 5.9 (c) 16.3 (d) 110.8
- 22** The horizontal range and maximum height of a projectile are equal. The angle of projection is → CBSE AIPMT 2012
 (a) $\theta = \tan^{-1}\left(\frac{1}{4}\right)$ (b) $\theta = \tan^{-1}(4)$
 (c) $\theta = \tan^{-1}(2)$ (d) $\theta = 45^\circ$
- 23** A missile is fired for maximum range with an initial velocity of 20 m/s. If $g = 10 \text{ m/s}^2$, the range of missile is → CBSE AIPMT 2011
 (a) 50 m (b) 60 m
 (c) 20 m (d) 40 m
- 24** A particle of mass m is projected with a velocity v making an angle of 45° with the horizontal. The magnitude of angular momentum of projectile about the point of projection when the particle is at its maximum height h is
 (a) zero (b) $\frac{mvh}{\sqrt{2}}$
 (c) $\frac{mvh^2}{\sqrt{2}}$ (d) None of these

DAY PRACTICE SESSION 2

PROGRESSIVE QUESTIONS EXERCISE

- 1** The sum of the magnitudes of two forces acting at a point is 16 N. The resultant of these forces is perpendicular to the smaller forces and has a magnitude of 8 N. If the smaller forces of magnitude x , then the value of x is
 (a) 2 N (b) 4 N (c) 6 N (d) 7 N
- 2** If the magnitude of sum of two vectors is equal to the magnitude of difference of the two vectors, the angle between these vectors is → NEET 2016, CBSE AIPMT 1991
 (a) 90° (b) 45° (c) 180° (d) 0°
- 3** The value of n so that vectors $2\hat{i} + 3\hat{j} - 2\hat{k}$, $5\hat{i} + n\hat{j} + \hat{k}$ and $-\hat{i} + 2\hat{j} + 3\hat{k}$ may be coplanar, will be
 (a) 18 (b) 28 (c) 9 (d) 36
- 4** A projectile is given an initial velocity of $(i + 2j)$ m/s, when i is along the ground and j is along the vertical. If $g = 10 \text{ m/s}^2$, the equation of its trajectory is
 (a) $y = x - 5x^2$ (b) $y = 2x - 5x^2$
 (c) $4y = 2x - 5x^2$ (d) $4y = 2x - 25x^2$
- 5** A police jeep is chasing a culprit going on a motorbike. The motorbike crosses a turning at a speed of 72 km/h. The jeep follows it at a speed of 90 km/h, crossing the turning 10 s later than the bike. Assuming that they travel at constant speeds, how far from the turning will the jeep catch up with the bike? (in km)
 (a) 1 (b) 2 (c) 3 (d) 4
- 6** A boat takes 2 h to travel 8 km and back in still water. If the velocity of water 4 km/h, the time taken for going up stream 8 km and coming back is
 (a) 2 h (b) 2 h 40 min (c) 1 h 20 min
 (d) Cannot be estimated with the information given
- 7** A man wants to reach point B on the opposite bank of a river flowing at a speed as shown in figure. What minimum speed relative to water should the man have, so that he can reach point B?
 (a) $u\sqrt{2}$ (b) $u/\sqrt{2}$ (c) $2u$ (d) $u/2$
- 8** A particle starting from the origin (0, 0) moves in a straight line in the XY-plane. Its coordinates at a later time are $(\sqrt{3}, 3)$. The path of the particle makes with the X-axis an angle of
 (a) 30° (b) 45° (c) 60° (d) 0°
- 9** A ball is rolled off along the edge of the table with horizontal velocity 4 m/s. It hits the ground after time 0.4 s. Which of the following statement is wrong. ($g = 10 \text{ m/s}^2$)
 (a) The height of table is 0.8 m.
 (b) It hits the ground at an angle of 60° with the vertical.
 (c) It covers a horizontal distance 1.6 m from the table.
 (d) It hits the ground with vertical velocity 4 m/s.



- 10** A ship A is moving Westwards with a speed of 10 km/h and ship B 100 km South of A , is moving Northwards with a speed of 10 km/h. The time after which the distance between them becomes shortest, is → CBSE AIPMT 2015
 (a) 0 h (b) 5 h (c) $5\sqrt{2}$ h (d) $10\sqrt{2}$ h

- 11** Two particles A and B , move with constant velocities $\mathbf{v}_1$ and $\mathbf{v}_2$. At the initial moment, their position vectors are $\mathbf{r}_1$ and $\mathbf{r}_2$ respectively. The condition for particles A and B for their collision is → CBSE AIPMT 2015
 (a) $\frac{\mathbf{r}_1 - \mathbf{r}_2}{|\mathbf{r}_1 - \mathbf{r}_2|} = \frac{\mathbf{v}_2 - \mathbf{v}_1}{|\mathbf{v}_2 - \mathbf{v}_1|}$ (b) $\mathbf{r}_1 \cdot \mathbf{v}_1 = \mathbf{r}_2 \cdot \mathbf{v}_2$
 (c) $\mathbf{r}_1 \times \mathbf{v}_1 = \mathbf{r}_2 \times \mathbf{v}_2$ (d) $\mathbf{r}_1 - \mathbf{r}_2 = \mathbf{v}_1 - \mathbf{v}_2$

- 12** The position vector of a particle $\mathbf{R}$ as a function of time is given by $\mathbf{R} = 4 \sin(2\pi t) \hat{\mathbf{i}} + 4 \cos(2\pi t) \hat{\mathbf{j}}$ where R is in metre, t is in seconds and $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ denote unit vectors along x and y -directions, respectively. Which one of the following statements is wrong for the motion of particle? → CBSE AIPMT 2015
 (a) Acceleration is along $-\mathbf{R}$
 (b) Magnitude of acceleration vector is $\frac{v^2}{R}$, where v is the velocity of particle
 (c) Magnitude of the velocity of particle is 8 m/s
 (d) Path of the particle is a circle of radius 4 m

ANSWERS

SESSION 1	1 (b)	2 (b)	3 (d)	4 (d)	5 (a)	6 (a)	7 (a)	8 (b)	9 (b)	10 (a)
	11 (c)	12 (a)	13 (c)	14 (c)	15 (c)	16 (b)	17 (d)	18 (d)	19 (c)	20 (c)
	21 (a)	22 (b)	23 (d)	24 (b)						
SESSION 2	1 (c)	2 (a)	3 (a)	4 (b)	5 (a)	6 (b)	7 (b)	8 (c)	9 (b)	10 (b)
	11 (a)	12 (c)								

Hints and Explanations

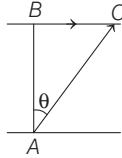
SESSION 1

- 1** A scalar quantity has same value for observers with different orientation of the axes. Since, value of scalar is independent of the direction of its observation.
- 2** Given, $\mathbf{R} = \mathbf{A} + \mathbf{B}$
 $\therefore R^2 = R^2 + R^2 + 2\mathbf{A} \cdot \mathbf{B} \cos \theta$
 or $\cos \theta = -\frac{1}{2}$
 $\therefore \theta = 120^\circ$
- 3** A vector parallel to $\mathbf{A}$ will be $n\mathbf{A}$ or $(3n\hat{\mathbf{i}} + 4n\hat{\mathbf{j}})$
 Now, $|n\mathbf{A}| = |\mathbf{B}|$ is given
 Hence,
 $n\sqrt{9 + 16} = \sqrt{49 + 576}$
 or $n = 5$
 $\therefore n\mathbf{A} = 15\hat{\mathbf{i}} + 20\hat{\mathbf{j}}$
- 4** When two non-zero vectors are represented by the two adjacent sides of a parallelogram, then the resultant is given by the diagonal of the parallelogram passing through the point of intersection of the two vectors $\mathbf{b} + \mathbf{e} = \mathbf{f}$.

- 5** Component of $\mathbf{A}$ along $\hat{\mathbf{i}} + \hat{\mathbf{j}}$
 $\Rightarrow \mathbf{A} \cdot \hat{\mathbf{B}} = \mathbf{A} \cdot \frac{\mathbf{B}}{|\mathbf{B}|} = \frac{(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}}) \cdot (\hat{\mathbf{i}} + \hat{\mathbf{j}})}{\sqrt{2}} = \frac{5}{\sqrt{2}}$
- 6** Given, $|\mathbf{A} \times \mathbf{B}| = \sqrt{3} (\mathbf{A} \cdot \mathbf{B})$
 $\Rightarrow AB \sin \theta = \sqrt{3} AB \cos \theta$
 $\Rightarrow \tan \theta = \sqrt{3} \Rightarrow \theta = 60^\circ$
- 7** $\mathbf{A} \times \mathbf{B} = (4\hat{\mathbf{i}} + 6\hat{\mathbf{j}}) \times (2\hat{\mathbf{i}} + 3\hat{\mathbf{j}})$
 $= 12(\hat{\mathbf{i}} \times \hat{\mathbf{j}}) + 12(\hat{\mathbf{j}} \times \hat{\mathbf{i}})$
 $= 12(\hat{\mathbf{i}} \times \hat{\mathbf{j}}) - 12(\hat{\mathbf{i}} \times \hat{\mathbf{j}}) = 0$
 Again, $\mathbf{A} \cdot \mathbf{B} = (4\hat{\mathbf{i}} + 6\hat{\mathbf{j}}) \cdot (2\hat{\mathbf{i}} + 3\hat{\mathbf{j}})$
 $= 8 + 18 = 26$
 Again, $\frac{|\mathbf{A}|}{|\mathbf{B}|} = \frac{\sqrt{16 + 36}}{\sqrt{4 + 9}} \neq \frac{1}{2}$
 Also, $\mathbf{B} = \frac{1}{2} \mathbf{A}$
 $\Rightarrow \mathbf{A}$ and $\mathbf{B}$ are parallel and not anti-parallel.
- 8** $\mathbf{A} \cdot \mathbf{B} = AB \cos \theta$
 Given, $\mathbf{A} = 4\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - 4\hat{\mathbf{k}}, \mathbf{B} = 3\hat{\mathbf{i}} + \hat{\mathbf{j}} + 4\hat{\mathbf{k}}$
 $\Rightarrow \mathbf{A} \cdot \mathbf{B} = (4\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - 4\hat{\mathbf{k}}) \cdot (3\hat{\mathbf{i}} + \hat{\mathbf{j}} + 4\hat{\mathbf{k}})$
 $= 4 \times 3 + 4 - 16 = 0$
 $\Rightarrow \mathbf{A} \cdot \mathbf{B} = 0 \Rightarrow \cos \theta = 0 \Rightarrow \theta = 90^\circ$

- 9** The coefficients of $\mathbf{i}, \mathbf{j}, \mathbf{k}$ should be a constant ratio.
 or $\frac{2}{-4} = \frac{3}{-6} = \frac{1}{\lambda}$ or $\lambda = -2$
- 10** Given, $\mathbf{A} \cdot \mathbf{B} = \mathbf{A} \times \mathbf{B}$
 $\Rightarrow AB \cos \theta = AB \sin \theta \Rightarrow \cos \theta = \sin \theta$
 $\Rightarrow \tan \theta = 1 \Rightarrow \theta = 45^\circ$
- 11** Let, $\mathbf{a} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 8\hat{\mathbf{k}},$
 $\mathbf{b} = 4\hat{\mathbf{j}} - 4\hat{\mathbf{i}} + \alpha\hat{\mathbf{k}} = -4\hat{\mathbf{i}} + 4\hat{\mathbf{j}} + \alpha\hat{\mathbf{k}}$
 Given, $\mathbf{a} \perp \mathbf{b} \Rightarrow \mathbf{a} \cdot \mathbf{b} = 0$
 $\Rightarrow (2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 8\hat{\mathbf{k}}) \cdot (-4\hat{\mathbf{i}} + 4\hat{\mathbf{j}} + \alpha\hat{\mathbf{k}}) = 0$
 $\Rightarrow -8 + 12 + 8\alpha = 0 \Rightarrow 8\alpha = -4$
 $\Rightarrow \alpha = -\frac{1}{2}$
- 12** Resultant, $R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$
 Given, $R = P\sqrt{10}, A = 2P, B = \sqrt{2}P$
 $\therefore P\sqrt{10} = \sqrt{4P^2 + 2P^2 + 4\sqrt{2}P^2 \cos \theta}$
 $\Rightarrow P\sqrt{10} = \sqrt{6P^2 + 4\sqrt{2}P^2 \cos \theta}$
 On, squaring both sides, we have
 $10P^2 = 6P^2 + 4\sqrt{2}P^2 \cos \theta$
 $4P^2 = 4\sqrt{2}P^2 \cos \theta$
 $\Rightarrow \cos \theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$

- 13** Given, AB = Velocity of boat = 8 km/h
 AC = Resultant velocity of boat = 10 km/h



$$\therefore BC = \text{Velocity of river} = \sqrt{AC^2 - AB^2}$$

$$= \sqrt{(10)^2 - (8)^2} = 6 \text{ km/h}$$

- 14** From the figure, the x -component remain unchanged, while the y -component is reverse. Then, the velocity at point B is $(2\hat{i} - 3\hat{j})$ m/s.

- 15** Given, $x = 5t - 2t^2$

Velocity of the particle,

$$v_x = \frac{dx}{dt} = \frac{d}{dt}(5t - 2t^2) = 5 - 4t$$

Acceleration, $a_x = \frac{dv_x}{dt} = -4 \text{ ms}^{-2}$

Also, $y = 10t$

Velocity, $v_y = \frac{dy}{dt} = 10$

$\therefore$ Acceleration, $a_y = \frac{dv_y}{dt} = 0$

$\therefore$ Net acceleration of the particle,

$$\mathbf{a}_{\text{net}} = a_x \hat{i} + a_y \hat{j} = (-4 \text{ ms}^{-2}) \hat{i}$$

or $\mathbf{a}_{\text{net}} = -4 \hat{i} \text{ ms}^{-2}$

- 16** Given, initial velocity (u) = $3\hat{i} + 4\hat{j}$

Final velocity (v) = ?

Acceleration (a) = $(0.4 \hat{i} + 0.3 \hat{j})$

Time (t) = 10 s

From first equation of motion,

$$\mathbf{v} = \mathbf{u} + \mathbf{a}t$$

$$\mathbf{v} = 3\hat{i} + 4\hat{j} + 10(0.4\hat{i} + 0.3\hat{j})$$

$$\mathbf{v} = 7\hat{i} + 7\hat{j} \Rightarrow |\mathbf{v}| = 7\sqrt{2}$$

- 17** Velocity, $\mathbf{v}_{\text{av}} = \frac{(x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j}}{t_2 - t_1}$

$$= \frac{(13 - 2)\hat{i} + (14 - 3)\hat{j}}{5 - 0}$$

$$= \frac{11\hat{i} + 11\hat{j}}{5} = \frac{11}{5}(\hat{i} + \hat{j})$$

- 18** $T = \frac{2u_y}{g}$, $H = \frac{u_y^2}{2g}$ and $R = u_x T$

When a horizontal acceleration is also given to the projectile u_y , T and H will remain unchanged while the range will become

$$R' = u_x T + \frac{1}{2} a T^2$$

$$= R + \frac{1}{2} \frac{g}{4} \left(\frac{4u_y^2}{g^2} \right) = R + H$$

and maximum height will be H .

- 19** Horizontal component of velocity at angle 60° = Horizontal component of velocity at 45°

$$\text{i.e. } u \cos 60^\circ = v \cos 45^\circ$$

$$\text{or } 147 \times \frac{1}{2} = v \times \frac{1}{\sqrt{2}} \text{ or } v = \frac{147}{\sqrt{2}} \text{ m/s}$$

Vertical component of

$$u_y = u \sin 60^\circ = \frac{147\sqrt{3}}{2} \text{ m}$$

Vertical component of

$$v_y = v \sin 45^\circ = \frac{147}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = \frac{147}{2} \text{ m}$$

$$\text{but } v_y = u_y + at$$

$$\therefore \frac{147}{2} = \frac{147\sqrt{3}}{2} - 9.8t \text{ or } t = 5.49 \text{ s}$$

- 20** Velocity $v = v_0 + gt + at^2$

$$\frac{dx}{dt} = v_0 + gt + at^2$$

Integrate on both sides,

$$\int dx = \int v_0 dt + \int gtdt + \int at^2 dt$$

$$x = v_0 t + \frac{1}{2} gt^2 + \frac{at^3}{3} + C$$

Given, $x = 0$ at $t = 0$

$$\therefore C = 0$$

$$x = v_0 t + \frac{1}{2} gt^2 + \frac{1}{3} at^3$$

$$\text{At } t = 1 \text{ second, } x = v_0 + \frac{1}{2} g + \frac{1}{3} a$$

- 21** $y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$

For equatorial trajectories for same angle of projection

$$\frac{8}{u^2} = \text{constant}$$

$$\Rightarrow \frac{9.8}{5^2} = \frac{g'}{3^2}$$

$$g' = \frac{9.8 \times 9}{25} = 3.528 \text{ m/s}^2$$

$$= 3.5 \text{ m/s}^2$$

- 22** Given, Range (R) = maximum height (H)

$$\text{Also, } R = \frac{u^2 (2\sin\theta \cos\theta)}{g}, H = \frac{u^2 \sin^2 \theta}{2g}$$

$$\therefore \frac{u^2 (2\sin\theta \cos\theta)}{g} = \frac{u^2 \sin^2 \theta}{2g}$$

$$\Rightarrow 2\cos\theta = \frac{\sin\theta}{2}$$

$$\Rightarrow \tan\theta = 4$$

$$\Rightarrow \theta = \tan^{-1}(4)$$

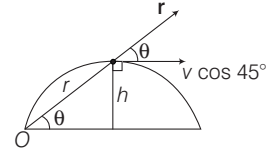
- 23** Maximum range of projectile is given by

$$R_{\text{max}} = \frac{u^2}{g}$$

Given, $u = 20 \text{ m/s}$ and $g = 10 \text{ m/s}^2$

$$\therefore R_{\text{max}} = \frac{(20)^2}{10} = \frac{400}{10} = 40 \text{ m}$$

- 24** The angular momentum of a particle is given by



$$\mathbf{L} = \mathbf{r} \times \mathbf{m} \mathbf{v}$$

$$L = mvr \sin \theta$$

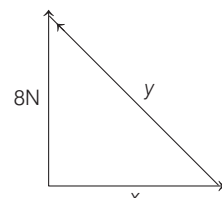
From figure,

$$L = r m (v \cos 45^\circ) \sin \theta$$

$$= \frac{mv}{\sqrt{2}} (r \sin \theta) = \frac{mvh}{\sqrt{2}}$$

SESSION 2

- 1** Given, $x + y = 16$



$$\text{Also, } y^2 = 8^2 + x^2$$

$$\text{or } y^2 = 64 + (16 - y)^2 \quad [\because x = 16 - y]$$

$$\text{or } y^2 = 64 + 256 + y^2 - 32y$$

$$\text{or } 32y = 320 \text{ or } y = 10 \text{ N}$$

$$\therefore x + 10 = 16 \text{ or } x = 6 \text{ N}$$

- 2** Suppose two vectors are P and Q .

It is given that $|\mathbf{P} + \mathbf{Q}| = |\mathbf{P} - \mathbf{Q}|$

Let angle between P and Q is ϕ .

$$\therefore P^2 + Q^2 + 2PQ \cos \phi = P^2 + Q^2 - 2PQ \cos \phi$$

$$\Rightarrow 4PQ \cos \phi = 0$$

$$\Rightarrow \cos \phi = 0 \quad [\because P, Q \neq 0]$$

$$\Rightarrow \phi = \frac{\pi}{2} = 90^\circ$$

- 3** For given vectors to be coplanar,

$$\mathbf{A} \times \mathbf{B} \times \mathbf{C} = 0$$

$$\mathbf{A} = 2\hat{i} + 3\hat{j} - 2\hat{k} \Rightarrow \mathbf{B} = 5\hat{i} + n\hat{j} + \hat{k}$$

$$\therefore \mathbf{C} = -\hat{i} + 2\hat{j} + 3\hat{k}$$

$$\begin{vmatrix} 2 & 3 & -2 \\ 5 & n & 1 \\ -1 & 2 & 3 \end{vmatrix} = 0$$

$$\Rightarrow 2(3n - 2) - 3(15 + 1) - 2(10 + n) = 0$$

$$\Rightarrow 6n - 4 - 45 - 3 - 20 - 2n = 0$$

$$\Rightarrow 4n = 72, n = 18$$

- 4** The equation of trajectory of a particle, fired, with an initial velocity u at an angle of projection θ ,

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$$

$$= x \tan \theta - \frac{gx^2}{2u^2} \sec^2 \theta$$

$$= x \tan \theta - \frac{gx^2}{2u^2} (1 + \tan^2 \theta)$$

Now, magnitude of velocity vector
 $u = \hat{i} + 2\hat{j} \Rightarrow u = \sqrt{(1)^2 + (2)^2} = 5 \text{ m/s}$
 and angle of projection is given by

$$\tan \theta = \frac{\hat{j} \text{ component}}{\hat{i} \text{ component}} = \frac{2}{1} = 2$$

$$\tan \theta = 2$$

So, from eq (i), we have

$$y = 2x - \frac{10 \times x^2}{2 \times 5} (1 + 4) = 2x - 5x^2$$

5 $v_p = 90 \text{ km/h} = 25 \text{ m/s}$

$$v_c = 72 \text{ km/h} = 20 \text{ m/s}$$

In 10 s culprits reach point B from A.

Distance covered by culprit,

$$S = vt = 20 \times 10 = 200 \text{ m}$$

At time $t = 10 \text{ s}$, the police jeep is 200 m behind the culprit.

Relative velocity between jeep and culprit is $25 - 20 = 5 \text{ m/s}$

$$\text{Time} = \frac{S}{v} = \frac{200}{5} = 40 \text{ s}$$

[Relative velocity is considered]

In 40 s, the police jeep will move from A to a distance S

$$\text{where, } S = vt = 25 \times 40 = 1000 \text{ m} = 1 \text{ km away}$$

The jeep will catch up with the bike 1 km far from the turning.

6 Boat covers distance of 16 km in a still water in 2 h

$$\text{i.e. } v_B = \frac{16}{2} = 8 \text{ km/h}$$

Now, velocity of water $v_w = 4 \text{ km/h}$

Time taken for going upstream

$$t_1 = \frac{8}{v_B - v_w} = \frac{8}{8 - 4} = 2 \text{ h}$$

As water current oppose the motion of boat, therefore time taken for going downstream

$$t_2 = \frac{8}{v_G + v_w} = \frac{8}{8 + 4} = \frac{8}{12} \text{ h}$$

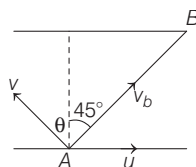
[water current helps the motion of boat]

$$\therefore \text{Total time} = t_1 + t_2$$

$$= \left(2 + \frac{8}{12} \right) \text{ h}$$

$$= 2 \text{ h } 40 \text{ min}$$

7 Let v be the speed of boatman in still water,



Resultant of v and u should be along AB . Components of v_b (absolute velocity of boatman) along x and y directions are,

$$v_x = u - v \sin \theta$$

$$\text{and } v_y = v \cos \theta$$

$$\text{Further, } \tan 45^\circ = \frac{v_y}{v_x}$$

$$\text{or } 1 = \frac{v \cos \theta}{u - v \sin \theta}$$

$$v = \frac{u}{\sin \theta + \cos \theta}$$

$$= \frac{u}{\sqrt{2} \sin (\theta + 45^\circ)}$$

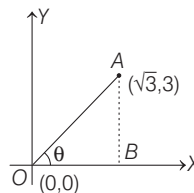
v is minimum at,

$$\theta + 45^\circ = 90^\circ \text{ or } \theta = 45^\circ$$

$$\text{and } v_{\min} = \frac{u}{\sqrt{2}}$$

8 Draw the situation as shown. OA represents the path of the particle starting from origin $O(0, 0)$. Draw a perpendicular from point A to X -axis. Let path of the particle makes an angle θ with the X -axis, then

$$\begin{aligned} \tan \theta &= \text{slope of line } OA \\ &= \frac{AB}{OB} = \frac{3}{\sqrt{3}} = \sqrt{3} \text{ or } \theta = 60^\circ \end{aligned}$$



9 Height of table

$$h = \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times (0.4)^2 = 0.8 \text{ m}$$

Horizontal distance covered $= u_x t$

$$= 4 \times 0.4 = 1.6 \text{ m}$$

Vertical velocity on reaching ground

$$v_y = u_y + a_y t = 0 + 10 \times 0.4 = 4 \text{ m/s}$$

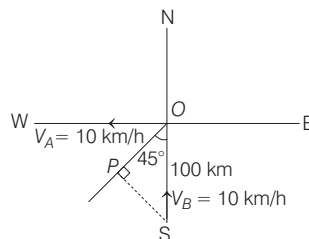
Horizontal velocity on reaching ground

$$v_c = u_x = 4 \text{ m/s}$$

If θ is the angle at which the ball hits the ground with the vertical, then

$$\tan \theta = \frac{v_x}{v_y} = \frac{4}{4} = 1 \Rightarrow \theta = 45^\circ$$

10



$$\sin 45^\circ = \frac{PS}{OS}$$

$$\Rightarrow \frac{1}{\sqrt{2}} = \frac{PS}{100}$$

$$PS = \frac{100}{\sqrt{2}} = \frac{100\sqrt{2}}{2} = 50\sqrt{2}$$

Relative velocity between A and B is

$$v_{BA} = \sqrt{v_A^2 + v_B^2} = 10\sqrt{2}$$

$$t = \frac{50\sqrt{2}}{10\sqrt{2}}$$

$$\Rightarrow t = 5 \text{ h}$$

11 For two particles A and B move with constant velocities v_1 and v_2 . Such that two particles to collide, the direction of the relative velocity of one with respect to other should be directed towards the relative position of the other particle.

$$\text{i.e. } \frac{\mathbf{r}_1 - \mathbf{r}_2}{|\mathbf{r}_1 - \mathbf{r}_2|} \rightarrow \text{direction of relative position of 1 w.r.t. 2.}$$

$$\text{Similarly, } \frac{\mathbf{v}_1 - \mathbf{v}_2}{|\mathbf{v}_1 - \mathbf{v}_2|} \rightarrow \text{direction of velocity of 2 w.r.t. 1.}$$

So, for collision of A and B, we get

$$\frac{\mathbf{r}_1 - \mathbf{r}_2}{|\mathbf{r}_1 - \mathbf{r}_2|} = \frac{\mathbf{v}_2 - \mathbf{v}_1}{|\mathbf{v}_2 - \mathbf{v}_1|}$$

12 (i) The position vector of a particle $\mathbf{R}$ as a function of time is given by

$$\mathbf{R} = 4 \sin (2\pi t) \hat{i} + 4 \cos (2\pi t) \hat{j}$$

x -component,

$$x = 4 \sin 2\pi t \quad \dots(i)$$

y -component,

$$y = 4 \cos 2\pi t \quad \dots(ii)$$

Squaring and adding both equations, we get

$$x^2 + y^2 = 4^2 [\sin^2 (2\pi t) + \cos^2 (2\pi t)]$$

i.e. $x^2 + y^2 = 4^2$ i.e. equation of circle and radius is 4 m.

(ii) Acceleration vector,

$$\mathbf{a} = \frac{v^2}{R} (-\hat{\mathbf{R}}), \text{ while } v \text{ is velocity of a particle.}$$

(iii) Magnitude of acceleration vector,

$$a = \frac{v^2}{R}$$

(iv) As, we have $v_x = +4 (\cos 2\pi t) 2\pi$ and $v_y = -4 (\sin 2\pi t) 2\pi$

Net resultant velocity,

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{(8\pi)^2 (\cos^2 2\pi t + \sin^2 2\pi t)}$$

$$v = 8\pi \quad [\because \cos^2 2\pi t + \sin^2 2\pi t = 1]$$

So, option (c) is incorrect.